Contents

Part A - Machine Learning

This part presents the overall findings and interpretation of the Machine Learning Project after comparing the original and updated results. The updated analysis was performed after adding the new six months of data to the existing dataset without retraining the models. The purpose of this part is to understand whether the original regression and classification models remained stable after the append process and whether the reusable machine learning system continued to produce consistent results on the larger unseen dataset.

The analysis begins with the regression model, followed by the Logistic Classification model, the Classification Confusion Matrix, the comparison of all classification models, and the Reliability Analysis. Finally, the overall findings of the Machine Learning Project are brought together to understand whether the original conclusions remained valid after adding the new data and whether the reusable machine learning framework continued to maintain its original behaviour.

1. Regression Model Performance

1.1 Purpose

The purpose of this section is to compare the original and updated regression model results after adding the new six months of data. The comparison helps in understanding whether the reusable regression model remained stable when evaluated on a larger unseen test dataset and whether the original regression conclusions continued to remain valid.

1.2 Output Fact Table

Source: error_summary_view

Set Type	Metric	Original	Updated
Test	MAE	**0.00256**	**0.002697**
Test	RMSE	**0.00342**	**0.003580**
Test	Nifty Standard Deviation	**0.00826**	**0.008595**

Test	MAE / Standard Deviation	**0.31000**	**0.313787**
Test	RMSE / Standard Deviation	**0.41000**	**0.416488**

1.3 Detailed Interpretation

MAE

The MAE increased only slightly after adding the new data. Even though the test period became larger, the average prediction error remained almost the same. This shows that the regression model was able to carry its learning to the new unseen data without losing its overall quality.

RMSE

RMSE also increased only by a very small amount. Since RMSE reacts more to larger errors, this shows that the additional six months of data did not introduce many extreme prediction errors, and the model continued to behave consistently.

Nifty Standard Deviation

The market itself became slightly more volatile after the append process. Therefore, a small increase in prediction error is expected, and the model should be understood in the context of the movement of the market rather than the error values alone.

MAE Relative to Market Volatility

Even after accounting for the increase in market volatility, the MAE remained almost at the same proportion of the market movement. This shows that the practical usefulness of the regression model remained stable after adding the new data.

RMSE Relative to Market Volatility

RMSE relative to market volatility also remained very close to the original result. This indicates that the model continued to control larger prediction errors well and did not become more sensitive after the append process.

1.4 Overall Conclusion

The regression results remained highly consistent after adding the new data. Although the test period became larger and the market volatility increased slightly, the prediction errors increased only proportionally. This shows that the original regression model remained both reusable and stable, and the original conclusions continued to hold.

2. Logistic Classification Performance

2.1 Purpose

The purpose of this section is to compare the original and updated Logistic Classification results after adding the new six months of data. The comparison helps in understanding whether the reusable classification model remained stable on a larger unseen test dataset and whether its overall predictive performance continued to remain consistent after the append process.

2.2 Output Fact Table

Source: classification_metrics_view

Set Type	**Metric**	**Original**	**Updated**
Test	Accuracy	**0.863**	**0.8635**
Test	Precision	**0.875**	**0.8784**
Test	Recall	**0.896**	**0.8627**

2.3 Detailed Interpretation

Accuracy

The test Accuracy remained almost identical after adding the new data. Even though the test dataset became larger, the model continued classifying unseen observations with almost the same overall performance. This shows that the classification model remained stable after the append process.

Precision

The Precision improved slightly after adding the new data. This means that when the model predicted a positive movement, it became slightly more reliable in making correct predictions. Although the improvement was small, it shows that the model did not lose its ability to identify positive observations accurately.

Recall

The Recall decreased after the append process, showing that the model became slightly more conservative in identifying positive observations. Instead of predicting more positive cases, the model avoided making unnecessary positive predictions, which helped maintain its overall classification performance.

2.4 Overall Conclusion

The classification model remained stable after adding the new data. The overall classification performance changed very little, showing that the reusable model continued to perform consistently on unseen observations. Although the Recall decreased slightly, the model maintained its overall predictive quality, and the original classification conclusions remained valid.

3. Classification Confusion Matrix

3.1 Purpose

The purpose of this section is to compare the original and updated confusion matrix after adding the new six months of data. The comparison helps in understanding whether the reusable classification model continued to identify both positive and negative market movements consistently on a larger unseen test dataset, while also evaluating whether the overall classification behaviour remained stable after the append process.

3.2 Output Fact Table

Source: classification_confusion_matrix_view

Set Type	Metric	Original	Updated
Test	True Positives (TP)	**240**	**289**
Test	True Negatives (TN)	**201**	**255**
Test	False Positives (FP)	**34**	**40**
Test	False Negatives (FN)	**36**	**46**
Test	Total Rows	**511**	**630**

3.3 Detailed Interpretation

True Positives (TP)

The number of correctly classified positive observations increased after adding the new data. Since the test dataset also became larger, this shows that the model continued identifying positive market movements successfully on the additional unseen observations.

True Negatives (TN)

The correctly classified negative observations also increased with the larger test dataset. This shows that the model maintained its ability to identify both positive and negative market movements after the append process.

False Positives (FP)

The False Positive predictions increased only slightly compared to the increase in the total test observations. This suggests that the model did not become noticeably more aggressive in making incorrect positive predictions after adding the new data.

False Negatives (FN)

The False Negative predictions also increased slightly as more unseen observations were added. This increase is expected because the overall test dataset became larger, and it does not indicate any major deterioration in the model's behaviour.

Test Dataset Size

The increase in the number of test observations came from the additional six months of appended market data. This allowed the reusable model to be evaluated on a larger unseen dataset without changing the original training process.

3.4 Overall Conclusion

The confusion matrix changed mainly because the test dataset became larger after the append process. Both the correct and incorrect classifications increased in proportion to the additional observations, showing that the overall behaviour of the classification model remained stable. The confusion matrix therefore supports the classification metrics and confirms that the reusable classification model continued to perform consistently after adding the new data.

4. All Model Performance

4.1 Purpose

In this section, the overall performance of all four machine learning models is compared after adding the new six months of data. The comparison helps in understanding whether the reusable classification models remained stable on a larger unseen test dataset and whether any one model could be clearly identified as better than the others based on the overall evaluation metrics.

4.2 Output Fact Table

Source: ensemble_metrics_view

Model	Set Type	Metric	Original	Updated
Logistic	Test	Accuracy	**0.863**	**0.8635**
Logistic	Test	Precision	**0.875**	**0.8785**
Logistic	Test	Recall	**0.896**	**0.8627**
Tree	Test	Accuracy	**0.847**	**0.8540**
Tree	Test	Precision	**0.863**	**0.8384**
Tree	Test	Recall	**0.891**	**0.8985**
Controlled Ensemble	Test	Accuracy	**0.831**	**0.8302**
Controlled Ensemble	Test	Precision	**0.834**	**0.8314**
Controlled Ensemble	Test	Recall	**0.858**	**0.8537**
Unrestricted Ensemble	Test	Accuracy	**0.826**	**0.8286**
Unrestricted Ensemble	Test	Precision	**0.830**	**0.8328**

Unrestricted Ensemble	Test	Recall	**0.851**	**0.8478**

4.3 Detailed Interpretation

Logistic Model

The Logistic model remained highly stable after adding the new data. The Accuracy remained almost identical, while the Precision improved slightly and the Recall reduced slightly after the append process. Overall, these small changes did not affect the overall classification behaviour, showing that the Logistic model continued to perform consistently on the larger unseen dataset.

Tree Model

The Tree model also remained stable after adding the new observations. Although the Precision reduced slightly, both the Accuracy and Recall improved after the append process. This shows that the Tree model continued classifying unseen observations effectively, with only small changes in its overall performance.

Controlled Ensemble Model

The Controlled Ensemble model showed very little change after adding the new data. The Accuracy, Precision and Recall all remained very close to their original values, indicating that the model continued performing consistently after the append process without any meaningful deterioration in its classification behaviour.

Unrestricted Ensemble Model

The Unrestricted Ensemble model also remained highly consistent after adding the new observations. The Accuracy and Precision improved slightly, while the Recall decreased only marginally. This shows that the overall classification performance remained stable, and the additional unseen data did not significantly change the behaviour of the model.

Overall Model Comparison

All four classification models remained highly stable after adding the new data. The changes in Accuracy, Precision and Recall were generally small across all the models, showing that the reusable classification system continued to perform consistently on unseen observations. The overall differences between the models also remained very small, making it difficult to distinguish one model as significantly better than the others based only on these evaluation metrics. This shows that the Reliability Analysis becomes the most important comparison for understanding the overall stability and practical behaviour of the models.

4.4 Overall Conclusion

Overall, all four machine learning models remained stable after the append process, with only small changes observed in their overall evaluation metrics. The updated results continued to support the original understanding that the reusable classification models were able to generalise well on the larger unseen dataset. Since the overall performance remained very similar across all the models, the Reliability Analysis becomes the key factor for identifying the most suitable model for long-term practical use.

5. Reliability Analysis

5.1 Purpose

In this section, the Reliability Analysis compares how the confidence of each machine learning model changed after moving from the training data to the larger unseen test dataset. Unlike the traditional evaluation metrics, this analysis helps in understanding how the models behaved internally after the append process and whether their confidence continued to remain reliable on new unseen observations.

5.2 Output Fact Table

Source: reliability_analysis_view

Model	**Confidence Category**	**Train**	**Test**	**Change**
Logistic	High Confidence Correct	**73.1%**	**70.0%**	**-3.1%**
Logistic	High Confidence Wrong	**6.9%**	**5.0%**	**-1.9%**
Logistic	Low Confidence Correct	**12.8%**	**15.9%**	**+3.1%**
Logistic	Low Confidence Wrong	**7.2%**	**8.5%**	**+1.3%**
Tree	High Confidence Correct	**73.5%**	**68.4%**	**-5.1%**
Tree	High Confidence Wrong	**4.2%**	**5.4%**	**+1.2%**
Tree	Low Confidence Correct	**15.1%**	**16.5%**	**+1.4%**
Tree	Low Confidence Wrong	**7.2%**	**9.6%**	**+2.4%**

Controlled Ensemble	High Confidence Correct	**75.9%**	**70.8%**	**-5.1%**
Controlled Ensemble	High Confidence Wrong	**5.2%**	**6.7%**	**+1.5%**
Controlled Ensemble	Low Confidence Correct	**11.4%**	**12.2%**	**+0.8%**
Controlled Ensemble	Low Confidence Wrong	**7.5%**	**9.7%**	**+2.2%**
Unrestricted Ensemble	High Confidence Correct	**80.2%**	**73.8%**	**-6.4%**
Unrestricted Ensemble	High Confidence Wrong	**4.3%**	**7.0%**	**+2.7%**
Unrestricted Ensemble	Low Confidence Correct	**8.1%**	**8.9%**	**+0.8%**
Unrestricted Ensemble	Low Confidence Wrong	**7.4%**	**10.0%**	**+2.6%**

5.3 Detailed Interpretation

Logistic Model

The Logistic model behaved differently from all the other models after moving from the training data to the unseen test data. Although the High Confidence Correct predictions reduced slightly, the High Confidence Wrong predictions also reduced after the append process. Most of the reduction shifted towards the Low Confidence Correct category, showing that the model became more careful rather than more confidently wrong. This shows that the Logistic model adjusted to the new data without losing its overall understanding of the dataset. A model that becomes slightly more cautious instead of more confidently wrong is much more reliable for future unseen data, making the Logistic model the most stable choice in this project.

Tree Model

The Tree model also lost some High Confidence Correct predictions after moving to the unseen data. However, unlike the Logistic model, a larger proportion of those predictions shifted towards the wrong prediction categories. This shows that the model became less careful on unseen observations and was more likely to make incorrect predictions with confidence. Although the overall evaluation metrics remained good, the Reliability Analysis shows that the Tree model became less dependable than the Logistic model after the append process.

Controlled Ensemble Model

The Controlled Ensemble model followed a similar pattern to the Tree model. As confidence reduced on the unseen data, a larger proportion of the predictions shifted towards the wrong prediction categories rather than remaining correct. This shows that some of the confidence learned during the training period did not transfer effectively to the larger unseen dataset. Although the model remained stable based on the traditional evaluation metrics, its confidence became less reliable after the append process.

Unrestricted Ensemble Model

The Unrestricted Ensemble model continued to produce the highest proportion of High Confidence Correct predictions, showing that it remained the strongest-performing model based on the traditional evaluation metrics. However, it also experienced the largest reduction in High Confidence Correct predictions after moving to the unseen data. A larger proportion of this confidence shifted towards the wrong prediction categories, showing that the model became more aggressive when classifying observations it had never seen before. This does not make the model poor, but it makes it less reliable than the Logistic model for long-term practical use.

5.4 Overall Interpretation

The traditional evaluation metrics such as Accuracy, Precision and Recall showed that all the models remained stable after adding the new data. However, these metrics only evaluate the final predictions and do not explain how the models behaved internally after moving from the training data to unseen observations. The Reliability Analysis answers that question. Every model lost some High Confidence Correct predictions after the append process, which is expected because unseen data is naturally more difficult. The important difference is where those predictions moved. The Logistic model mainly shifted towards the Low Confidence Correct category, meaning it became more careful while still remaining correct. The Tree and Ensemble models shifted a larger proportion of their confidence towards the wrong prediction categories, showing that they became more aggressive on unseen data. This difference becomes important for long-term practical use because a model that becomes slightly more cautious is safer than a model that becomes more confidently wrong. This shows that the Logistic model remained the most stable and reliable model in this project, making the Reliability Analysis the most important evaluation for distinguishing the behaviour of the reusable machine learning models.

5.5 Overall Conclusion

Overall, the Reliability Analysis provided the clearest understanding of how the machine learning models behaved after adding the new six months of data. Although all four models remained stable based on their overall evaluation metrics, the movement of confidence between the training and unseen test data showed clear differences in their long-term reliability. This shows that the Logistic model remained the most suitable model for the reusable system because it became more cautious rather than more confidently wrong after the append process, while the other models transferred a larger proportion of their confidence towards incorrect predictions.

6. Overall Machine Learning Findings

6.1 Purpose

In this section, the overall findings of the Machine Learning Project are brought together after comparing the original and updated results. The purpose is to understand whether the original conclusions remained valid after adding the new six months of data and whether the reusable machine learning system continued to remain stable after the append process.

6.2 Overall Machine Learning Findings

Regression Performance

The regression model remained stable after adding the new data. Although the prediction errors increased slightly, the increase remained proportional to the increase in market volatility. This shows that the model continued to perform in almost the same way on the updated data, and the original understanding of the regression model remained the same.

Classification Performance

The Logistic, Tree and Ensemble models all continued to perform consistently after adding the new data. The overall evaluation metrics such as Accuracy, Precision and Recall remained almost unchanged across all the models, making it difficult to distinguish one model from another based only on these traditional evaluation metrics. This shows that the reusable classification system continued to perform well on the larger unseen test dataset without any major deterioration in its overall predictive performance.

Reliability Analysis

The Reliability Analysis became the most important part of the classification evaluation because it explained something that Accuracy, Precision and Recall could not show. While all the models performed well, the Logistic model transferred more of its confidence towards the Low Confidence Correct category instead of the wrong prediction categories. The Tree and Ensemble models transferred a larger proportion of their confidence towards incorrect predictions after moving to the unseen data. This showed that the Logistic model remained the most stable model after adding the new data.

Model Comparison

The Controlled and Unrestricted Ensemble models continued to produce the best overall classification performance after adding the new data. However, the Reliability Analysis showed that both models also experienced a larger shift towards the wrong prediction categories after moving to the unseen data. This shows that the best-performing model is not always the most stable model when new data is added. The comparison also showed that selecting a model should not depend only on the traditional evaluation metrics, but also on how the model behaves after moving to unseen data.

Reusable Machine Learning System

The reusable machine learning system successfully processed the additional six months of data without retraining the models. More importantly, the updated results remained almost the same as the original results. This shows that the reusable system is not only able to process new data but is also able to maintain the original behaviour of the models after the append process.

6.3 Overall Conclusion

The append analysis showed that the Machine Learning Project remained stable after adding the new data. The original understanding of the regression and classification models remained the same, while the Reliability Analysis gave a better understanding of how the models behaved after moving to unseen data. Overall, the reusable machine learning system continued to produce consistent results, giving confidence that it can continue to work as new data is added in the future.

PART B - Natural Language Processing

This part presents the overall findings and interpretation of the Natural Language Processing Project after comparing the original and updated results. The updated analysis was performed after adding the new six months of financial news data to the existing dataset using the same reusable NLP framework. The purpose of this part is to understand whether the original NLP signals remained valid after the append process and whether the reusable NLP system continued to produce meaningful results on the larger and more diverse news dataset.

The analysis begins with Sentiment Analysis, followed by Entity Analysis, Theme Analysis, the Combined Context Signal, the Earnings and Theme Signal Distribution, and the Earnings and Theme Signal vs Stock and Market Analysis. Finally, the overall findings of the Natural Language Processing Project are brought together to understand whether the original conclusions remained valid after adding the new data and whether the reusable NLP framework continued to maintain its original analytical behaviour.

1. Sentiment Analysis

1.1 Purpose

In this section, the original and updated Sentiment Analysis results are compared after adding the new six months of financial news data. The purpose is to understand whether the Sentiment signal became stronger after analysing a much larger news dataset and whether the original understanding of the Sentiment signal remained valid after the append process.

1.2 Output Fact Table

Source: sentiment_summary

Metric	Original	Updated
Total Days	**149**	**268**
Match Count	**75**	**136**
Failure Count	**74**	**132**
Match Percentage	**50.34%**	**50.75%**
Failure Percentage	**49.66%**	**49.25%**

1.3 Detailed Interpretation

Total News Days

The number of news days increased significantly after adding the new data. This provided a much larger dataset to evaluate whether the Sentiment signal became stronger after analysing more financial news.

Match Count

The number of matching days increased because more news observations were added. This increase was expected as the dataset became much larger after the append process.

Failure Count

The number of mismatching days also increased with the larger dataset. This shows that simply adding more news did not reduce the number of days where the Sentiment signal and market movement disagreed.

Match Percentage

Although the Match Count increased, the Match Percentage remained almost the same. This shows that the additional news data did not improve the overall strength of the Sentiment signal.

Failure Percentage

The Failure Percentage also remained almost unchanged after adding the new data. This further shows that the behaviour of the Sentiment signal remained almost the same even with a much larger dataset.

1.4 Overall Conclusion

The overall behaviour of the Sentiment signal remained almost unchanged after adding the new data. The larger dataset increased both the matching and mismatching observations in almost the same proportion, keeping the Match and Failure percentages close to 50%. This shows that the limitation was not the amount of news available but the Sentiment signal itself. Simply adding more data did not make the Sentiment signal stronger for explaining market movement, and the original conclusion of the NLP Project remained the same.

2. Entity Analysis

2.1 Purpose

In this section, the original and updated Entity Analysis results are compared after adding the new six months of financial news data. The purpose is to understand whether the Entity signal remained useful after the append process and whether the original understanding of the Entity Analysis continued to remain valid on the larger news dataset.

2.2 Detailed Interpretation

High Signal Entities

The updated dataset identified many more High Signal entities than before. This happened because much more news data was available after adding the new observations. As more news was collected, more companies, government events, organisations and financial terms crossed the signal threshold. This shows that Entity Analysis naturally expands as the amount of news increases.

Strongest Entities

The strongest entities were no longer limited to a few companies. The updated results included entities such as Union Budget, Wikipedia, Tata, AMD, Trump, OpenAI, Axis Bank and Sensex. This shows that the important entities changed with the current market environment and the latest news. The important entities are therefore not fixed and will continue to change whenever new market events occur.

Entity Diversity

The updated Entity Analysis covered a much wider range of entities than the original project. Instead of focusing on only a few entities, it identified companies, government events, financial institutions, technology companies, political events and market indices together. This gives a broader understanding of what topics were important during the updated time period, but it also shows that the market cannot be explained by only a few important entities.

Reusable NLP Framework

The filtering process remained exactly the same after adding the new data. Even though the important entities changed completely, the same reusable framework was able to identify them automatically without changing the logic of the project. This shows that the reusable NLP system continued to work successfully after the append process.

Limitation of Entity Analysis

The updated results showed that the important entities changed significantly after adding the new data. This is expected because the financial market is influenced by new companies, events, government decisions and economic developments every day. As a result, Entity Analysis keeps changing with the news, making it difficult to use as a stable standalone signal for explaining market movement over different time periods.

2.3 Overall Conclusion

The updated results further strengthened the original conclusion of the NLP Project. Entity Analysis helps in understanding who or what the market is talking about, but it does not explain why the market moved. Since the important entities continue to change as new news is added, Entity Analysis remained a supporting signal rather than a strong standalone signal for explaining stock returns. This is also one of the reasons why Theme Analysis became the strongest NLP signal in the project.

3. Theme Analysis

3.1 Purpose

In this section, the original and updated Theme Analysis results are compared after adding the new six months of financial news data. The purpose is to understand whether the Theme signal continued to remain the strongest NLP signal after the append process and whether the original understanding of Theme Analysis remained valid on the larger news dataset.

3.2 Detailed Interpretation

Strong Theme Signals

The updated analysis identified more strong themes than the original project because the larger news dataset captured a much wider range of market events. Instead of only a few dominant themes, the updated analysis identified Banking & Credit, Digital & Technology, Corporate & Industry Development, Risk & Uncertainty, Economic Growth, Earnings Performance, and AI & Emerging Technology as the strongest themes. This shows that Theme Analysis continued to capture broader market narratives as more news was added.

Positive and Negative Themes

The market environment changed after adding the new data. While the original analysis showed several themes with positive market behaviour, the updated analysis contained more uncertain and negative market events, resulting in most themes showing negative average returns. Risk & Uncertainty continued to remain the most negative theme, while Economic Growth became the only theme with a positive average return. This shows that the change was mainly driven by the nature of the updated news rather than a weakness of the Theme Analysis itself.

Theme Diversity

The updated Theme Analysis expanded beyond earnings, market sentiment and technology to include banking, corporate development, economic growth, artificial intelligence and risk. The larger dataset allowed Theme Analysis to capture a much wider range of market situations than before. This provided a better understanding of the different factors influencing the market instead of focusing on only a few areas.

Theme Quality

Theme Analysis continued to reduce the bias observed in Entity Analysis by grouping similar phrases into broader market narratives. Even though more themes appeared after adding the new data, they continued to represent overall market situations instead of individual companies or entities. This shows that Theme Analysis remained a more meaningful way of understanding market behaviour than analysing individual entities alone.

Reusable NLP Framework

The same reusable NLP framework continued to identify meaningful themes after adding the new data without changing the overall analytical process. Although the additional news changed the themes that were identified, the same framework continued to produce meaningful results while maintaining the original project logic. This shows that the reusable NLP system remained stable after the append process.

3.3 Overall Conclusion

The updated results continued to support the original conclusion of the NLP Project. Although the larger dataset made the market behaviour more diverse and more difficult to interpret because it contained a much wider variety of news, Theme Analysis continued to explain the market better than Sentiment Analysis and Entity Analysis by focusing on the broader meaning of the news rather than individual words or entities. This shows that the original understanding of the project remained valid, and Theme Analysis continued to remain the strongest standalone NLP signal.

4. Combined Context Signal

4.1 Purpose

In this section, the original and updated Combined Context Signal results are compared after adding the new six months of financial news data. The purpose is to understand whether combining multiple NLP signals produced a stronger market signal after the append process and whether the original understanding of the combined signal remained valid on the larger news dataset.

4.2 Output Fact Table

Source: combined_vs_market_accuracy

Metric	Original	Updated
Match Percentage	**44.0%**	**50.22%**
Mismatch Percentage	**56.0%**	**49.78%**

4.3 Detailed Interpretation

Match Percentage

The Match Percentage improved after adding the new data, increasing from 44.00% to 50.22%. Although the updated results showed some improvement, the combined signal remained close to 50%. This shows that combining multiple NLP signals did not produce a consistently stronger market signal, even after analysing a much larger news dataset.

Mismatch Percentage

The Mismatch Percentage reduced after adding the new data, decreasing from 56.00% to 49.78%. This shows that the combined signal became slightly better after the append process. However, the improvement was not large enough to change the overall understanding of the combined signal.

Combined Signal Behaviour

Although more news data was available after adding the new observations, combining Sentiment Analysis, Entity Analysis and Theme Analysis did not improve the overall market signal. Instead, the additional signals reduced the strength of Theme Analysis by introducing more variation and noise into the final combined score. This shows that adding more signals does not always produce a stronger signal, especially when the strongest standalone signal is already capturing most of the useful market information.

Methodology of Combining Signals

The same methodology was used after adding the new data. Even though the updated results improved slightly, the overall behaviour remained almost the same. This shows that the limitation was not only the dataset but also the methodology used to combine the different NLP signals. Since each signal represents a different aspect of the news, combining them using a simple weighting and averaging process was not sufficient to capture the overall market behaviour.

Reusable NLP Framework

The reusable NLP framework successfully generated the updated combined signal after adding the new data without changing the overall analytical process. Although the numerical results changed slightly, the same framework continued to generate the combined signal while maintaining the original project logic. This shows that the reusable NLP system continued to work successfully after the append process.

4.4 Overall Conclusion

The updated analysis further strengthened the original conclusion of the NLP Project. Simply combining multiple NLP signals did not necessarily produce a better market signal. Theme Analysis already captured most of the useful market information, while adding Sentiment Analysis and Entity Analysis introduced additional noise that reduced the overall effectiveness of the combined signal. This shows that a well-designed standalone signal can be more useful than a complex multi-signal approach, and the original understanding of the project remained valid after adding the new data.

5. Earnings and Theme Signal Distribution

5.1 Purpose

In this section, the original and updated distribution of the combined Earnings and Theme signals is compared after adding the new six months of financial news data. The purpose is to understand how the relationship between company earnings and broader market themes changed after the append process and whether the original understanding of the combined signal remained valid on the larger news dataset.

5.2 Output Fact Table

Source: theme_signal_distribution

Combined Signal	**Original**	**Updated**
Strong Positive	**82.5%**	**4.43%**
Conflicting Positive	**15.0%**	**87.34%**
Strong Negative	**2.5%**	**8.23%**

5.3 Detailed Interpretation

Strong Positive Signals

The number of Strong Positive signals reduced significantly after adding the new data, decreasing from 82.50% to 4.43%. This does not mean that the earnings signal became weaker. Instead, the updated dataset contained a much wider range of market themes, many of which were negative, reducing the number of situations where both company earnings and market themes moved in the same positive direction.

Conflicting Positive Signals

Conflicting Positive became the dominant category after adding the new data, increasing from 15.00% to 87.34%. This shows that company earnings continued to remain mostly positive, while the broader market themes became much more negative. This difference reflects the gap between the company's own communication and the overall market environment rather than a weakness in the analytical framework.

Strong Negative Signals

Strong Negative signals increased after the append process, rising from 2.50% to 8.23%. This shows that weak earnings together with negative market themes appeared more frequently in the updated period because the larger dataset captured a wider range of uncertain market conditions.

Signal Distribution

The larger news dataset changed the overall distribution of the combined signals. The market themes became more diverse and more negative, while the earnings signal continued to remain largely positive. This naturally shifted most observations towards the Conflicting Positive category.

Understanding the Results

The updated results further explain the difference between the two signals. Earnings represent the company's own view of its performance, while Theme Analysis represents the broader market environment. As more market news was added, this difference became much clearer, leading to a much higher proportion of Conflicting Positive signals.

5.4 Overall Conclusion

The append analysis showed that combining earnings and themes continued to provide useful information for understanding company-level behaviour. However, the results also showed that company communication and broader market conditions do not always move together. This further strengthened the original conclusion that earnings and themes should be understood as two different perspectives rather than expecting them to produce one strong combined market signal.

6. Earnings and Theme Signal vs Market Analysis

6.1 Purpose

In this section, the original and updated results of the combined Earnings and Theme signal are compared against both stock-level and market-level behaviour after adding the new six months of financial news data. The purpose is to understand whether the combined signal became more useful after the append process and whether it explained individual company behaviour better than the overall market.

6.2 Output Fact Table

Source: theme_signal_vs_market_analysis

Type	Metric	Original	Updated
Stock	Match Percentage	68.97%	72.37%
Stock	Mismatch Percentage	31.03%	27.63%
Market	Match Percentage	51.72%	59.21%
Market	Mismatch Percentage	48.28%	40.79%

6.3 Detailed Interpretation

Stock Alignment

The Stock Match Percentage increased after adding the new data, improving from 68.97% to 72.37%, while the Mismatch Percentage reduced from 31.03% to 27.63%. This shows that combining company earnings with market themes continued to explain individual company behaviour effectively. The larger dataset further strengthened this relationship after the append process.

Market Alignment

The Market Match Percentage also improved after adding the new data, increasing from 51.72% to 59.21%, while the Mismatch Percentage reduced from 48.28% to 40.79%. However, the

Market Match Percentage remained noticeably lower than the Stock Match Percentage. This shows that overall market movement depends on many additional factors beyond company earnings and market themes.

Stock vs Market Behaviour

The combined signal continued to explain stock behaviour better than overall market behaviour. The same pattern remained after adding the new data, with Stock Alignment continuing to stay clearly higher than Market Alignment. This shows that company earnings and market themes naturally explain the behaviour of individual companies better than the behaviour of the overall market because the market is influenced by many additional economic, political and global factors.

Understanding the Combined Signal

The additional news data improved both the stock-level and market-level results, but it did not change the main purpose of the combined signal. The combined Earnings and Theme signal remained much more useful for understanding company-level behaviour than for explaining overall market movement. This shows that the strength of the signal continued to remain at the company level after the append process.

6.4 Overall Conclusion

The append analysis further strengthened the original conclusion of the NLP Project. Combining Earnings and Theme signals provides a useful way of understanding individual company behaviour, but it is not sufficient to explain the overall market by itself. The updated results confirmed that company-level signals naturally perform better at the company level than at the broader market level, and the original understanding of the project remained valid after adding the new data.

7. Overall Natural Language Processing Findings

7.1 Purpose

In this section, the overall findings of the Natural Language Processing Project are brought together after comparing the original and updated results. The purpose is to understand whether the original conclusions remained valid after adding the new six months of financial news data and whether the reusable NLP framework continued to remain stable after the append process.

7.2 Overall Natural Language Processing Findings

Sentiment Analysis

The Sentiment signal remained almost the same after adding the new data. Even though the number of news observations increased significantly, the Match and Failure percentages remained close to 50%. This shows that simply adding more news did not make the Sentiment signal stronger for explaining market movement, and the original understanding of the Sentiment signal remained the same.

Entity Analysis

The important entities changed after adding the new data because the market itself changed over time. However, Entity Analysis continued to explain what the market was talking about rather than why the market moved. This shows that Entity Analysis remained a supporting signal instead of a strong standalone signal for explaining stock returns.

Theme Analysis

Theme Analysis continued to be the strongest NLP signal after adding the new data. Although the updated dataset introduced more themes and changed their distribution, Theme Analysis continued to explain the broader market behaviour better than Sentiment Analysis and Entity Analysis because it focused on the overall meaning of the news rather than individual words or entities.

Combined Context Signal

Combining Sentiment Analysis, Entity Analysis and Theme Analysis showed only a small improvement after adding the new data. However, the combined signal still could not perform better than Theme Analysis alone. This shows that simply combining multiple NLP signals does not always produce a stronger market signal, especially when the strongest standalone signal is combined with weaker supporting signals.

Earnings and Theme Signal

The combined Earnings and Theme signal continued to explain stock behaviour better than overall market behaviour after adding the new data. Although both the stock-level and market-level results improved after the append process, the stock results remained much stronger. This shows that company-level signals are naturally more useful for understanding individual company behaviour than the behaviour of the overall market.

Reusable NLP Framework

The reusable NLP framework continued to work successfully after the append process without changing the original analytical process. Although the larger news dataset changed the individual signals and introduced a wider variety of market events, the same analytical framework continued to generate meaningful results. This shows that the reusable NLP system remained stable while processing the additional news data.

7.3 Overall Conclusion

The append analysis showed that the Natural Language Processing Project remained stable after adding the new data. Although the individual signals changed because the updated dataset contained much more news and a wider variety of market events, the original understanding of the project remained the same. Sentiment Analysis continued to remain a weak signal, Entity Analysis remained a supporting signal, Theme Analysis continued to remain the strongest standalone NLP signal, and the combined Earnings and Theme signal continued to explain stock behaviour better than overall market behaviour. Overall, the reusable NLP framework continued to produce meaningful results without changing the original analytical process, giving confidence that the same framework can continue to remain useful as new financial news is added in the future.

PART C - Overall Understanding of the Four Projects

The first two parts of this document focused on the updated understanding of the Machine Learning and Natural Language Processing Projects individually. Each project was analysed separately to understand how the additional data affected its analytical behaviour and whether the original conclusions remained stable after the append process. This helped validate that both reusable analytical systems continued to produce meaningful results without changing their overall understanding.

However, analysing both projects separately does not provide the complete picture. Throughout the complete journey, these projects were never intended to remain independent. Every project solved a different problem, answered a different analytical question, and gradually became the foundation for the next stage of the work. Together, they formed one reusable analytical framework for understanding structured numerical data and unstructured financial information.

Therefore, this part moves beyond the individual project analysis and focuses on the complete understanding developed from all four projects together. It explains the common understanding shared by both projects, the differences between their analytical approaches, their practical applications, and finally the overall understanding developed from building the complete reusable analytical framework.

Rather than evaluating the individual results again, the objective of this part is to bring together the complete learning, understand how the different projects complement each other, and explain the overall contribution of the reusable analytical framework after successfully completing all four projects.

1. Common Understanding from Both Projects

1.1 Purpose of this Section

The Machine Learning and Natural Language Processing projects were built using different types of data, different analytical methods, and different workflows. Even though both projects worked differently, there were many common understandings that came from them after completing the reusable systems analysis. These common understandings are not based on one result or one model. Instead, they came from looking at the overall behaviour of both projects after adding the new data. This section explains the major understanding that both projects gave together and what they teach about building reusable analytical systems.

1.2 Both Projects Successfully Demonstrated Reusability

One of the common understandings that came from both projects is that both of them successfully remained reusable after adding the new data. The way both projects process the data is completely different, but after the append process, both of them continued to work without changing their overall workflow.

In the ML Project, the existing models generated the updated results without retraining the models. Similarly, in the NLP Project, the same reusable framework generated the updated signals after adding the new financial news. Although the internal processing was different, both projects continued to produce the updated results successfully.

This shows that reusability does not depend on using the same analytical methods. A reusable system can work with both structured and unstructured data as long as the overall framework continues to process the new data correctly.

1.3 Stability Was More Important Than Reusability

Another common understanding that came from both projects is that making a project reusable is not enough. The project should also continue to produce stable analytical results after adding the new data. If the results change completely every time new data is added, then there is very little practical use in making the project reusable.

This can be seen in both projects. In the ML Project, the regression and classification models remained stable after adding the new data, while the Reliability Analysis showed that the Logistic model continued to remain the most stable model. Similarly, in the NLP Project, many values changed after adding the new news data, but the overall understanding of the project remained the same. Theme Analysis continued to remain the strongest signal, while Sentiment and Entity Analysis continued to behave as supporting signals.

This shows that the real success of a reusable system is not only that it can process new data, but that it can continue to provide stable analytical understanding after the data changes.

1.4 More Data Did Not Automatically Produce Better Results

One of the most interesting understandings from both projects is that adding more data did not automatically improve the analytical results.

In the ML Project, six months of new market data was added, but the overall behaviour of the models remained almost the same. The prediction errors increased only slightly, and the classification models continued to perform consistently.

Similarly, in the NLP Project, a much larger amount of financial news was added. This increased the amount of information available, but it did not make every NLP signal stronger. Sentiment Analysis remained close to 50%, while the Combined Context Signal also showed only a small improvement. The biggest improvement was seen in Theme Analysis because the larger news dataset captured a wider range of market themes.

This shows that the quality and nature of the data are often more important than simply increasing the quantity of the data.

1.5 Simple Understanding Was More Valuable Than Complex Analysis

Another common understanding that came from both projects is that increasing the complexity of the analysis did not always produce a better understanding of the market.

In the ML Project, the Ensemble models produced the strongest overall performance, but the Reliability Analysis showed that the Logistic model remained the most stable model when evaluated on unseen data. Similarly, in the NLP Project, combining different NLP signals together did not produce a stronger signal than Theme Analysis alone. Theme Analysis continued to provide the best understanding because it focused on the overall meaning of the financial news instead of looking at individual words or entities.

This shows that a more complex analytical approach is not always better. Sometimes a simpler approach provides a clearer and more reliable understanding of the market.

1.6 The Original Understanding Remained Valid

Another important understanding from both projects is that the original conclusions remained valid after adding the new data.

Many numerical values changed in both projects because the datasets became larger. Some signals became stronger, some became weaker, and many observations changed. However, the overall understanding of both projects remained almost the same.

In the ML Project, the models remained stable, and the Reliability Analysis continued to identify the Logistic model as the most stable model. Similarly, in the NLP Project, the important themes and entities changed with the new news, but Theme Analysis continued to remain the strongest NLP signal.

This shows that the append process did not change the original understanding of either project. Instead, it strengthened the confidence that the reusable analytical framework continued to work as expected.

1.7 Both Projects Focused More on Understanding Than Prediction

One of the biggest understandings that came from completing both projects is that the final objective was not prediction alone. Instead, both projects became more focused on understanding the market.

The ML Project helped explain how the market behaved by analysing historical numerical relationships and checking whether those relationships remained stable after adding the new data. The NLP Project helped explain what was happening in the market by analysing financial news, themes, entities, and the overall meaning of the information.

This shows that both projects focused on improving the understanding of the market from different perspectives instead of trying to maximise prediction alone.

1.8 Overall Common Understanding

After completing both projects, one common understanding became very clear. Although the Machine Learning and Natural Language Processing projects used different data, different workflows, and different analytical methods, both of them reached very similar overall conclusions. Both projects remained reusable after adding the new data, both continued to produce stable analytical results, and both preserved their original understanding after the append process.

Another important understanding is that reusability alone is not enough. A reusable system should also continue to provide stable and meaningful analytical results as new data is added. Otherwise, there is very little practical value in making the system reusable. Both projects successfully demonstrated this behaviour and showed that the reusable analytical framework remained both reusable and analytically stable.

Finally, both projects also showed that building more complex analytical systems does not always lead to a better understanding of the market. Understanding the behaviour of the models, the signals, and the data itself proved to be much more valuable than simply increasing the complexity of the analytical methods. This became one of the biggest common understandings developed from both projects.

2. Different Understanding from Both Projects

2.1 Purpose of this Section

Although both the Machine Learning and Natural Language Processing projects successfully demonstrated reusable analytical systems, they were developed to understand different parts of the financial market. Both projects started with different objectives, worked on different types of data, followed different analytical approaches, and produced different types of understanding. These differences should not be seen as limitations of either project. Instead, they explain why both projects were required in this reusable analytical framework. This section explains the major differences between both projects and the unique understanding that each project contributed.

2.2 Both Projects Started with Different Foundations

One of the biggest differences between both projects is the foundation on which they were built.

The ML Project started with structured numerical data such as stock prices, returns, and market movements. The main objective was to understand whether historical numerical relationships remained stable after adding the new data and whether the trained models could continue to perform consistently on unseen observations.

On the other hand, the NLP Project started with unstructured financial news. Before any analysis could begin, the news first had to be converted into structured information using different NLP techniques. The main objective was to understand whether financial news could provide meaningful signals for explaining market behaviour.

This shows that both projects started from completely different foundations. One project started with numerical relationships, while the other started with financial information and market context.

2.3 Both Projects Helped Understand Different Parts of the Financial Market

Another important difference is the type of understanding that both projects provided.

The ML Project helped understand how the financial market behaved by analysing historical numerical relationships. It explained how prices and returns moved over time and whether these relationships remained stable after adding the new data.

The NLP Project helped understand what was happening in the market by analysing financial news. It explained the meaning, context, themes, and important events that influenced market behaviour during different time periods.

This shows that both projects explained different parts of the same financial market. The ML Project explained the numerical behaviour of the market, while the NLP Project explained the informational behaviour of the market.

2.4 The Behaviour of Both Projects Was Different After Adding the New Data

Another important difference appeared after the append process.

In the ML Project, adding six months of new market data produced only small changes in the overall results. Most of the regression and classification metrics remained stable, showing that the trained models continued to generalise well on the larger unseen dataset.

However, the NLP Project behaved differently. The larger news dataset introduced many new entities, new themes, and new financial events. Because of this, many individual NLP signals changed after adding the new data. Even though the individual values changed, the overall understanding of the project remained the same.

This shows that numerical data usually changes gradually over time, while financial news changes continuously because new events keep entering the market. Therefore, both projects naturally behaved differently after the append process.

2.5 The Most Important Understanding Was Different in Both Projects

One of the biggest differences between both projects came from their most important findings.

In the ML Project, the Reliability Analysis became the most important analytical finding. The traditional evaluation metrics such as Accuracy, Precision, and Recall showed that all the models performed well. However, the Reliability Analysis explained how the confidence of the models changed after moving from the training data to the unseen test data. This completely changed the understanding of the classification models and showed that the Logistic model remained the most stable model for future unseen data.

In the NLP Project, Theme Analysis became the most important analytical finding. Although Sentiment Analysis and Entity Analysis provided useful supporting information, Theme Analysis consistently remained the strongest NLP signal because it grouped the financial news into meaningful market themes instead of analysing individual words or entities separately.

This shows that the biggest understanding in both projects came from different analytical approaches. In the ML Project, the deeper understanding came from Reliability Analysis, while in the NLP Project, the deeper understanding came from Theme Analysis.

2.6 Both Projects Faced Different Analytical Challenges

Another important difference is the type of challenge faced by both projects.

The biggest challenge in the ML Project was understanding which model remained stable after adding the new data. The traditional evaluation metrics showed only small differences between the models, making it difficult to identify the most suitable model. The Reliability Analysis solved this problem by showing how the confidence of the models changed after moving to unseen observations.

The biggest challenge in the NLP Project was converting unstructured financial news into meaningful analytical signals. Simply analysing sentiment or important entities was not enough to explain market behaviour. The project gradually showed that understanding the overall theme of the news provided much better analytical understanding than analysing individual words or entities.

This shows that although both projects solved different problems, they both required deeper analysis before the final understanding could be reached.

2.7 Both Projects Required Different Reusable Architectures

Another difference can be seen in the reusable workflow itself.

The ML Project followed a highly structured workflow where most of the processing was performed directly inside BigQuery. Once the models were trained, the reusable system automatically generated the updated analytical results after adding the new data.

The NLP Project required a more modular workflow. Python was used for processing the financial news before the extracted information was transferred back into BigQuery for further analysis. Theme Analysis also required manual interpretation before the final themes could be generated.

This shows that the reusable architecture depended on the nature of the data being analysed. Structured numerical data and unstructured financial news required different reusable workflows to achieve the same overall objective.

2.8 Overall Different Understanding

After completing both projects, it became clear that the Machine Learning and Natural Language Processing projects were designed to understand different aspects of the financial market. The ML Project focused on understanding historical numerical behaviour, model stability, and prediction consistency, while the NLP Project focused on understanding financial information, market context, and the meaning behind the news.

Another important understanding is that the biggest contribution of both projects also came from different analytical findings. In the ML Project, the Reliability Analysis changed the understanding of the classification models by showing that stability is more important than looking at the traditional evaluation metrics alone. In the NLP Project, Theme Analysis changed the understanding of financial news by showing that grouping the news into broader themes provides much better analytical understanding than analysing individual words or entities.

Finally, these differences should not be seen as competition between both projects. Instead, they show that both projects explain different parts of the same financial market. The ML Project explains how the market behaves using structured numerical data, while the NLP Project explains the information and context behind that behaviour. Together, they provide a broader and more complete understanding of the financial market than either project could provide individually.

3. Practical Understanding of Both Projects

3.1 Purpose of this Section

After understanding the common and different learning from both projects, the next important question is whether these projects have any practical use in real financial analysis. Throughout the project, the main objective was never to build a system that predicts the market with complete accuracy. Instead, the objective was to build a reusable analytical framework that can continue to understand new financial data whenever it is added. This section explains where the complete framework can be useful in practice and what type of understanding it can provide.

3.2 The Framework Should Be Used for Understanding Rather Than Direct Prediction

One of the biggest practical understandings that came from this project is that the complete framework should not be seen as a direct prediction system. During the project, both the ML and NLP analyses showed that financial markets are influenced by many different factors, and no single model or signal can completely explain future market behaviour.

The ML Project showed that the models remained stable after adding the new data, but it also showed that higher model performance does not always mean better long-term stability. Similarly, the NLP Project showed that Theme Analysis explained market behaviour much better than Sentiment Analysis, but it also showed that financial news alone cannot explain every market movement.

This shows that the practical value of the framework is not predicting the future with certainty. Instead, it helps build a better understanding of why the market behaves the way it does.

3.3 The ML and NLP Projects Can Be Used Together to Build Better Market Understanding

Another important practical understanding is that both projects explain different parts of the same financial market.

The ML Project explains how the market behaves by analysing historical numerical relationships. It helps understand whether the market behaviour remains stable after new data is added and whether the analytical models continue to perform consistently.

The NLP Project explains what information is influencing the market by analysing financial news, market themes, important entities, and financial context. It provides an understanding of the information available during the same period.

This shows that both projects can be used together to build a broader understanding of the market. One explains the numerical behaviour of the market, while the other explains the information behind that behaviour.

3.4 The Framework Can Support Financial Analysis Rather Than Replace Human Decisions

Another important understanding from this project is that the framework should be used as a decision-support system rather than a decision-making system.

Throughout the project, many situations were observed where the analytical results required interpretation before any meaningful conclusion could be reached. For example, the Reliability Analysis was required to identify the most stable ML model, while Theme Analysis was required to identify the strongest NLP signal. Simply looking at the numerical outputs would not have produced the same understanding.

This shows that the framework is designed to support financial analysis by providing structured analytical understanding. The final decisions should still depend on human interpretation, market knowledge, and practical experience.

3.5 The Framework Can Continue to Work as New Data Is Added

One of the main objectives of the reusable systems project was to understand whether the analytical workflow could continue to work without rebuilding the complete project every time new data became available.

The append analysis showed that the ML Project continued to generate stable analytical results without retraining the models. Similarly, the NLP Project continued to generate updated signals using the same reusable workflow after adding the new financial news.

This shows that the framework can continue to support future financial analysis without redesigning the complete analytical process whenever new data becomes available.

3.6 The Framework Can Be Useful for Different Types of Financial Analysis

Another practical understanding is that the framework is flexible enough to support different types of financial analysis.

The ML Project can be used to analyse historical market behaviour, evaluate model stability, and understand the consistency of numerical relationships over time.

The NLP Project can be used to analyse financial news, understand important market themes, identify changing financial events, and explain the information available during different market situations.

Together, both projects provide a broader analytical understanding that can be useful for equity research, market monitoring, financial reporting, investment research, and academic financial analysis.

This shows that the framework is not limited to only one financial application. Instead, it provides a reusable analytical process that can support different types of financial studies.

3.7 The Biggest Practical Value of the Framework Is Its Analytical Reusability

One of the biggest practical understandings developed during this project is that the real value of the framework does not come from automation alone.

Many analytical systems can automatically process new data. However, if the analytical understanding changes completely every time new data is added, then the practical value of the system becomes very limited.

This project showed that both the ML and NLP workflows continued to produce meaningful analytical understanding after the append process. Even though many numerical values and individual signals changed, the overall understanding of both projects remained consistent.

This shows that the biggest practical contribution of the framework is its ability to remain analytically reusable as the data continues to evolve.

3.8 Overall Practical Understanding

After completing both projects, one practical understanding became very clear. The Machine Learning and Natural Language Processing projects should not be viewed as independent prediction systems. Instead, they should be viewed as complementary analytical systems that help understand different aspects of the financial market.

The ML Project explains the numerical behaviour of the market, while the NLP Project explains the information and context behind that behaviour. Together, they provide a broader understanding than either project could provide individually. More importantly, both projects demonstrated that the reusable analytical framework continued to work successfully after adding the new data while preserving the original analytical understanding.

Finally, the biggest practical contribution of this framework is not that it predicts the market with complete accuracy. Its biggest contribution is that it provides a reusable analytical process that can continue to generate stable and meaningful financial understanding as new structured and unstructured data becomes available. This makes the framework suitable as a decision-support and financial analysis tool rather than a fully automated decision-making system.

4. Overall Understanding of the Complete Reusable Analytical Framework

4.1 Purpose of this Section

After completing the Machine Learning Project, the Natural Language Processing Project, the Reusable Systems Project, and the final append analysis, the next important step is to understand the complete reusable analytical framework as one system instead of looking at every project separately. Each project answered a different question and solved a different problem, but together they formed one complete analytical workflow. This section explains the overall understanding that came from the complete framework and what it finally contributes after completing all four projects.

4.2 The Framework Was Built Step by Step

One of the biggest understandings from this complete journey is that the reusable analytical framework was not built in one stage. Every project solved a different problem, and each project became the foundation for the next one.

The Machine Learning Project helped understand the numerical behaviour of the financial market using structured data. After that, the Natural Language Processing Project expanded the understanding by analysing financial news and market information. The Reusable Systems Project then connected both projects into a reusable workflow so that new data could continue to move through the same analytical process. Finally, the append analysis validated whether the complete framework continued to work successfully after adding the new data.

This shows that the framework was developed gradually, where each project improved the understanding gained from the previous one.

4.3 The Framework Became More Than Individual Projects

Another important understanding is that the final framework became much bigger than the individual ML and NLP projects.

If both projects are studied separately, they explain only one part of the financial market. However, when they are viewed together inside the reusable system, they provide a much broader understanding. The numerical behaviour of the market, the financial information available in the market, and the reusable analytical workflow all become part of one complete system.

This shows that the complete framework should not be understood as a collection of independent projects. Instead, it should be understood as one analytical system where every project contributes a different part of the overall understanding.

4.4 Reusability Became the Central Understanding of the Framework

During the early stages of the project, the main focus was on understanding Machine Learning models and Natural Language Processing signals. However, after completing the Reusable Systems Project and the append analysis, a different understanding became much more important.

The project showed that creating models or extracting signals is only one part of the analytical process. The more important objective is whether the same analytical workflow can continue to work correctly after new data is added.

Both the ML and NLP projects successfully demonstrated this behaviour. Even though the datasets became larger and many observations changed, the overall analytical understanding remained stable.

This shows that reusability became the central understanding of the complete framework rather than only the individual analytical models.

4.5 The Framework Demonstrated Both Operational and Analytical Stability

Another important understanding is that the framework remained stable in two different ways.

The first was operational stability. The reusable workflow continued to process the new structured and unstructured data without changing the overall architecture.

The second was analytical stability. Even though many numerical values, themes, entities, and observations changed after the append process, the original conclusions of both projects remained valid.

This shows that the complete framework remained stable not only because it continued to execute successfully, but also because it continued to produce meaningful analytical understanding after adding the new data.

4.6 The Framework Helps Understand the Financial Market from Multiple Perspectives

Another important understanding is that the framework does not depend on only one type of financial analysis.

The Machine Learning Project explains the historical numerical behaviour of the market. The Natural Language Processing Project explains the financial information and context available during different market situations. Together, both projects help understand different parts of the same financial market.

This shows that the framework provides a broader analytical understanding than using only structured data or only financial news independently.

4.7 The Framework Should Continue to Evolve

One of the final understandings from this project is that the framework should not be considered complete forever.

As new market data and financial news continue to become available, the framework can continue to process the updated information using the same reusable workflow. The analytical conclusions should also continue to be validated whenever new data is added.

This shows that the framework is designed to evolve continuously while preserving its original analytical structure.

4.8 Overall Understanding of the Complete Reusable Analytical Framework

After completing all four projects, one overall understanding became very clear. The purpose of this complete work was never to build only a Machine Learning Project or only a Natural Language Processing Project. Instead, the purpose gradually evolved into building a reusable analytical framework that could continue to understand both structured numerical data and unstructured financial information using the same overall system.

Another important understanding is that the value of the framework does not come from any one individual model or signal. Instead, it comes from the combination of reusable workflows, stable analytical behaviour, structured numerical analysis, and financial information analysis. Each project contributed a different part towards this overall understanding, making the complete framework much stronger than any individual project.

Finally, the biggest contribution of this framework is that it successfully demonstrated how reusable analytical systems can continue to generate stable and meaningful understanding even after new data is added. Rather than focusing only on prediction or automation, the framework focuses on preserving analytical understanding as the data continues to evolve. This became the final understanding of the complete reusable analytical framework and the overall conclusion of the entire series of projects.

Part D - Project Journey and Learning Reflection

I still can't believe that I have completed this series of projects, especially because it was never supposed to happen this way. When I started this journey, I was not even sure where it would end up, as I had no idea it would eventually become a four-project series. At the beginning, I was simply focused on learning Artificial Intelligence. Very soon, I realised that to understand AI properly, I first had to understand Machine Learning and Natural Language Processing. My original plan was to build everything using Google Vertex AI, and I thought that once I provided the data, it would automatically build the models and allow me to focus on analysing the results.

But Google Vertex AI never worked on my Google Cloud account. I still don't know what the exact problem was, but I spent two or three days trying different solutions without any success. At one point, I had almost given up on learning Machine Learning altogether. I even considered joining an online course, something I had always tried to avoid because many of them are expensive and often focus more on theory than practical understanding. Python-based Machine Learning was always an option, but I did not want my learning journey to become completely code-oriented. I have never wanted to become a programmer. One of the things I like about AI is that it allows me to focus more on analysis while using it as an assistant for coding rather than making coding the centre of the work.

That was when BigQuery SQL entered the picture. At that time, I was not even sure whether it would help me, so I first started learning the basics of SQL. Surprisingly, I found SQL much easier to understand than Python. Then there was one moment when I finally started understanding the concept of SQL Views. It felt like one of those rare moments where everything suddenly started making sense. I had always built my projects in Excel, where they were dynamic in nature, but I could never control the complete workflow through proper checkpoints. With SQL Views, I realised that every major analytical decision could have its own checkpoint, and those checkpoints could be connected together to build one complete analytical workflow.

At that stage, I had never thought about reusability or reusable systems because that phase did not even exist in my mind. I also had no idea how the Machine Learning Project would finally be structured. But after understanding SQL Views, I completely changed the way I designed the project. Instead of building isolated queries, I started building connected analytical layers. Looking back now, I think this became one of the biggest turning points of the entire journey.

After that, I started building the Machine Learning Project using BigQuery SQL and SQL Views. The project was mentally exhausting, especially during the model evaluation stage, but every new model gradually added another level of understanding. I moved from Regression to Logistic Classification and finally to the Ensemble models. The transition between the models felt natural because every model extended the understanding developed from the previous one instead of replacing it. In the end, the complete Machine Learning Project was built using SQL Views as analytical checkpoints rather than relying on one large static structure.

When the NLP Project started, another unexpected challenge appeared. BigQuery SQL did not support the NLP functions that I needed. This time I was not as disappointed as before. I was definitely frustrated, but by then I had started believing that if one approach did not work, another solution probably existed. Using Python as an external processing layer created a completely different architecture for the NLP Project while still allowing BigQuery to remain the central analytical platform. Looking back now, I realise that this challenge helped me understand that different types of data require different analytical workflows, and that one solution cannot always be applied to every problem.

After completing both projects, I realised that the SQL Views I had created could do much more than simply organise queries. They could actually make the complete analytical workflow reusable. That understanding became the foundation of the Reusable Systems Project. The objective was no longer only to build Machine Learning and NLP projects, but to make both of them capable of processing new compatible data without rebuilding the complete workflow every time.

When I added the new data to the Machine Learning Project, the reusable workflow was tested for the first time with unseen compatible data. Instead of making me worried, the updated results actually gave me confidence that the project had been built correctly. The complete workflow remained connected, the analytical structure remained stable, and the updated results were generated successfully without changing the overall framework.

The NLP Project also brought its own set of challenges, especially because it worked with unstructured financial news instead of regular structured datasets. Some manual adjustments were still required, particularly in Theme Analysis, and those adjustments were reasonable because they were part of understanding language rather than simply processing numbers. Even with these challenges, the reusable workflow continued to work successfully, and the overall analytical understanding of the NLP Project remained stable after adding the new data.

In the end, the append analysis gave me much more confidence in the complete framework that I had built. The Machine Learning results remained highly stable, while the changes observed in the NLP Project were also understandable because financial news naturally changes over time. Going through every model and every NLP signal one step at a time helped me understand not only how they work, but also why they behave differently and what each of them contributes towards understanding the data.

Looking back now, I realise that if Google Vertex AI had worked on the very first day, I would probably never have discovered SQL Views, never built this reusable analytical framework, and never understood Machine Learning and Natural Language Processing in the way I do today. What first looked like a failure eventually became the biggest turning point of the entire journey.

More importantly, this journey taught me something beyond Machine Learning and Natural Language Processing. It taught me how to approach a completely new problem, understand the data, build an analytical workflow, interpret the results, and continuously improve the structure whenever I found a better solution. Today, I feel much more confident about approaching a completely new dataset because I now understand how to think through the analytical process instead of simply applying models.

Looking back now, I also realise that this project is not really an ending. It is more like the completion of one stage of my learning. The framework that I have built can continue to evolve whenever new market data becomes available, and many improvements can still be made in the future. But instead of starting from scratch every time, I now have a structured analytical foundation that I understand well. More than anything else, that gives me confidence about working on future analytical problems.

There is a quote that says, *"Sometimes the path that doesn't work becomes the reason you discover a much better one."* Looking back at this complete journey, I feel that this is exactly what happened to me. If one path had worked exactly as I expected, I might never have discovered a much better way of building projects. More than the models, the signals, or the reusable framework itself, this complete journey taught me how to think analytically, how to keep improving whenever I faced a challenge, and how every unexpected problem can become the beginning of something much better.